

Audio-Laya: Adapting a Pretrained Decision Model to Audio with a Lightweight Projector

A Technical Report on Robocall Screening, Semantic Alignment, and Transfer Evaluation

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Version 1.1 • September 2026

Abstract

This report studies a modular alternative to building one general-purpose model for every task: keep a pre-trained audio encoder frozen, connect its representations to a pretrained typed-decision model through a trainable projector, and adapt only the small set of parameters needed for a target decision. In Audio-Laya, a frozen Whisper encoder supplies audio features to Laya, a pretrained text decision model. On a 282-call spam-versus-legitimate test split, the projector-only system reached 99.6% Macro F1 and the projector-plus-decision-head system reached 100.0%, compared with 89.2% for a baseline that sends Whisper’s automatic speech recognition (ASR) transcript to Laya. However, spam labels are completely confounded with data source, so these scores do not establish real-world spam generalization. Direct-audio models transferred poorly to MInDS-14 intent classification; results on a 40-clip synthetic urgency set were variable and exploratory. An optional audio–transcript alignment objective substantially improved transcript retrieval but did not reliably improve new-task decisions. One three-epoch run took 492 seconds on an AMD Radeon RX 9070 XT. Overall, the evidence supports a proof of concept for task-specific modality adaptation—not a universal audio-understanding model. The trainable parameter counts are small relative to the frozen backbones; they are not the total model or deployment footprint.

Keywords: audio classification; modality adaptation; projector; frozen encoder; typed decisions; robocall screening; semantic alignment

Contents

1	Introduction	1
1.1	Plain-language summary	1
1.2	Research question and scope	1
2	Background and system architecture	2
2.1	Model variants	2
3	Methodology and evaluation protocol	3
3.1	Training and semantic alignment	3
3.2	Data and evaluation protocol	3
3.2.1	Robocall screening	3
3.2.2	Transfer datasets	3
3.2.3	Metrics	4
3.3	Reproducibility protocol and status	4
3.3.1	Data provenance and split construction	4
3.3.2	Models and training configuration	5
3.3.3	Environment, commands, and output artifacts	5
3.3.4	Checklist summary	6
4	Results and discussion	7
4.1	Quantitative results	7
4.1.1	Robocall test split	7
4.1.2	Zero-shot transfer	7
4.1.3	What semantic alignment changed	8
4.2	Interpretation	9
5	Limitations, responsible use, and conclusions	10
5.1	Limitations and responsible use	10
5.2	Conclusion	11
	References	12

Chapter 1

Introduction

1.1 Plain-language summary

Most general-purpose models try to handle many tasks in one large system. This project tests a different recipe: reuse a pretrained audio encoder, add a small trainable bridge (the *projector*), and connect the resulting audio features to an existing decision model. The intended output is a task-specific decision, such as *spam* or *legitimate*.

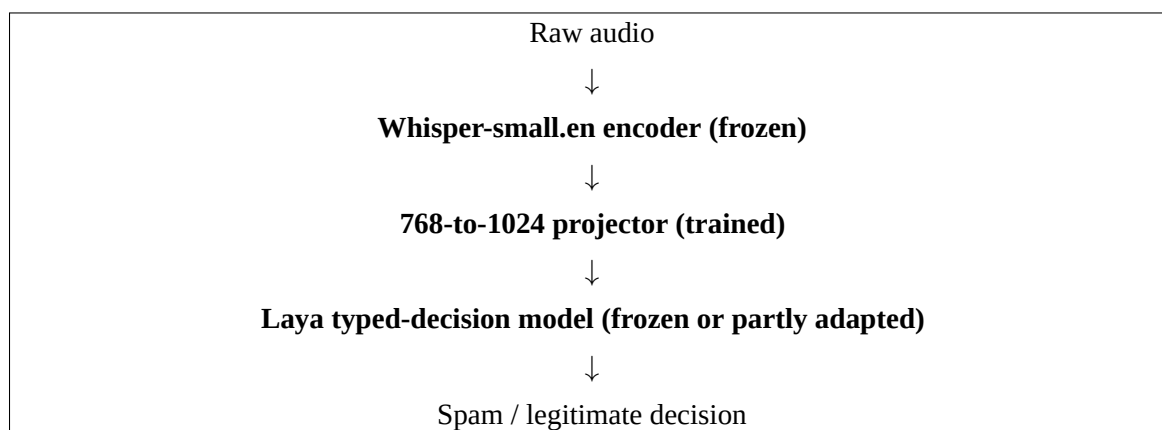
The experiment shows that this arrangement can score highly on one internal audio-classification split. It does *not* yet show that the system recognizes robocalls across new sources, understands arbitrary audio tasks, or transfers to vision and sensor data. The main test set has a serious source-label confound, and the transfer sets have substantial limitations. The honest conclusion is therefore: this is a promising modular adaptation recipe to investigate, not evidence of a universal model.

1.2 Research question and scope

The research question is whether audio features from a frozen, pretrained encoder can be made useful to a pretrained decision model by training a lightweight projector, optionally together with the decision layers. The project does not propose a new speech-recognition model and does not fine-tune the Whisper encoder.

The central design principle is:

Instead of building one model for every task, adapt pretrained perception encoders to small, domain-specific decision systems through lightweight trainable projectors.



The direct-audio path does not generate or consume transcripts at inference. In a separate semantic-alignment experiment, source transcripts are used as optional training supervision only. Only audio and transcript-text paths were evaluated; vision, video, and sensor inputs remain future directions.

Chapter 2

Background and system architecture

2.1 Model variants

The audio encoder is `openai/whisper-small.en`; the decision model is `convaiinnovations/laya` [1, 2]. A projector maps Whisper’s 768-dimensional features to the 1024-dimensional representation used by the Laya interface. The Laya checkpoint is a pretrained text decision model built on ModernBERT-large, not a newly trained speech model.

Table 2.1: What is updated in each experimental route. Parameter counts are trainable parameters, not the size of the full model loaded for inference.

Variant	Trainable components	Trainable parameters
A: projector only	Projector; Laya decision components remain frozen	1,838,592
B: projector + head	Projector, Laya type embedding, decision Transformer, and option scorer	28,086,785
C: ASR $\rightarrow$ Laya	No task-specific fine-tuning; Whisper transcribes, standard Laya processes text	0
A + semantic	Projector only; adds an audio–transcript alignment objective during training	1,838,592

Laya’s published model card describes a 421-million-parameter text model. Thus, “lightweight” here refers to the number of parameters updated and the measured adaptation run—not to the total deployed model size, memory requirement, or inference cost. The Whisper and Laya backbones remain part of the system. The Laya model card also describes text classification and spam-related use cases; this experiment therefore tests connecting audio to an existing decision capability, not learning the concept of spam from scratch [2].

Chapter 3

Methodology and evaluation protocol

3.1 Training and semantic alignment

The supervised task is binary classification of robocall audio as *spam* or *legitimate*. A and B use audio directly. C is a separate text route: Whisper ASR creates a transcript, then the unmodified Laya text interface makes a decision. These routes do not have matched inputs, so their scores should not be interpreted as a controlled comparison of audio and text models.

The semantic-alignment run used 22.91 hours of audio from the FTC and Harper training splits: 1,926 calls, 3,520 paired audio chunks, and 3,028 unique transcript texts. It trained the projector with

$$\mathcal{L} = \mathcal{L}_{\text{spam}} + 0.05 \mathcal{L}_{\text{InfoNCE}}, \quad \tau = 0.07, \quad (3.1)$$

where the contrastive term encourages an audio embedding to retrieve its paired transcript embedding. Whisper, the Laya text encoder, and the Laya decision head remain frozen. Transcripts are not used by the direct-audio inference path.

The recorded three-epoch semantic run took **492.11 seconds (8 minutes 12 seconds)** on an AMD Radeon RX 9070 XT. This wall time includes model loading, text-embedding preparation, training, and validation. A comparable wall-clock time was not recorded for the ten-epoch run; one measured run is not evidence that every training configuration finishes within a fixed time.

3.2 Data and evaluation protocol

3.2.1 Robocall screening

The internal test split contains 282 calls (576 audio chunks): 146 Harper Valley Bank calls labeled legitimate and 136 FTC robocall recordings labeled spam. Calls are averaged at the call level for the reported metrics. Although the split is held out from training, the source and label are perfectly aligned: all legitimate test calls come from Harper and all spam calls come from FTC. The model can therefore exploit source, recording, or conversational-style differences. The reported scores are pipeline results on this split, not an estimate of real-world spam-detection performance.

3.2.2 Transfer datasets

- **MinDS-14:** all 563 examples in the English-US training release, spanning 14 banking intents. This release has no official held-out test split. The full release is used here as a zero-shot scoring set; it is not used for gradient updates. The majority-class baseline is 8.5% accuracy and 1.1% Macro F1 [5, 6].

- **Urgency tone:** 40 English AI-generated speech clips, evenly split between urgent and non-urgent. The clips are synthetic and the scenario content may itself reveal the label. The majority baseline is 50.0% accuracy and 33.3% Macro F1 [7].

A and B were evaluated on the transfer tasks without gradient training on target-task labels. C uses text after ASR; for MInDS, a reference-transcript-to-Laya control is also reported. Neither C result is an audio-only comparison. The three-epoch semantic follow-up is an exception to a fully independent evaluation: it was requested after the earlier urgency result had been inspected, so the 40-clip set influenced the choice of epoch count.

3.2.3 Metrics

Accuracy is the fraction of correct decisions. Macro F1 averages the F1 score across classes, giving each class equal weight. AUROC summarizes how well a score ranks positive cases above negative cases across thresholds; for MInDS-14 we use one-vs-rest (OvR) macro AUROC. AUROC does not guarantee correct decisions at the default threshold and does not mean a probability is calibrated. Recall@ k measures whether the paired transcript appears among the top k retrieval candidates.

3.3 Reproducibility protocol and status

This section follows the relevant experimental-reproducibility, experimental-setting, statistical-uncertainty, compute, and asset-licensing categories in the NeurIPS Paper Checklist Guidelines [8]. It is checklist-aligned documentation, not a claim of formal NeurIPS submission compliance.

3.3.1 Data provenance and split construction

The source snapshots inspected for this report were:

- FTC Robocall Audio repository commit 5aa6f3bfa8563ce8c1c75ebf8a2271e6ff6b4272.
- HarperValleyBank repository commit 0bd721e877c4a85d8c13ff837e68661ea6200a98.
- MInDS-14 dataset revision 40ce77cb32a384e4d50a568e1ec39ac804019d33, pinned in its task configuration.
- Urgency-tone version 1, archive MD5 427b71b8019b0f7eebcccb162dc594e0.

The datasets have separate terms; this report does not redistribute recordings or transcripts. See source documentation and citations for attribution and licensing [3, 4, 5, 6, 7].

The FTC and Harper preparation scripts write the manifests used by training. Original calls are assigned to train/validation/test partitions at 80/10/10, stratified by class with seed 42; audio is chunked only after call-level splitting, so chunks from one call remain in one partition. The loader downmixes each chunk to mono and resamples to 16 kHz with polyphase resampling; chunks are at most 30 seconds. Training-only gain and noise augmentation are specified in the training configuration. The split protocol does not remove the source-label confound described above. Transfer-manifest acquisition and filtering notes are in `data/raw/external/minds14/README.md` and `data/raw/external/urgency-tone/README.md`.

3.3.2 Models and training configuration

The model identifiers are `openai/whisper-small.en` and `convaiinnovations/laya`. The default settings in `configs/poc.yaml` are 10 epochs, patience 3, batch size 1, two data workers, projector learning rate 10^{-4} , decision-head learning rate 10^{-5} , weight decay 0.01, gradient clipping at 1.0, seed 42, and mixed precision when supported. Training chunks are capped at 30 seconds; training-only augmentation uses a 3 dB gain and noise SNR sampled from 25–40 dB. The semantic runs add weight 0.05 and temperature 0.07; the three-epoch run overrides the configured epoch count. The seed initializes Python, NumPy, and PyTorch, but deterministic GPU kernels are not enforced.

The Hugging Face model identifiers are recorded, but their immutable weight revisions are not. A later download of the same identifiers may therefore resolve to different weights. Checkpoints and run records are present in the working copy under `runs/`; this report does not provide a public checkpoint archive.

3.3.3 Environment, commands, and output artifacts

The project README specifies Python 3.10 or newer and a PyTorch build suitable for the target hardware. `requirements.txt` uses minimum-version ranges and there is no lockfile. At report revision, the working environment reported Python 3.12.14, PyTorch 2.8.0+rocm7.1.0.lw.git7a520360, Transformers 5.17.0, NumPy 2.5.3, and scikit-learn 1.9.1; this is an observed environment snapshot, not a record tied to each historical run. The project root has no Git commit, and historical run records do not capture the full effective configuration, package snapshot, model revisions, or source-code revision. Thus the commands below provide a rerun path, but exact or bitwise reproduction is not established.

Run from the project root after obtaining the source data. The first block pins the two source repositories to the commits inspected for this report; install a hardware-matched PyTorch build before installing the remaining requirements.

```
python -m venv .venv
. .venv/bin/activate
# Install a hardware-matched PyTorch build in this environment first.
python -m pip install -r requirements.txt

git clone https://github.com/wspr-ncsu/robocall-audio-dataset.git data/raw/ftc
git -C data/raw/ftc checkout 5aa6f3bfa8563ce8c1c75ebf8a2271e6ff6b4272
git clone https://github.com/cricketclub/gridspace-stanford-harper-valley.git \
    data/raw/harper
git -C data/raw/harper checkout 0bd721e877c4a85d8c13ff837e68661ea6200a98
python -m src.data.prepare_ftc --root data/raw/ftc
python -m src.data.prepare_harper --root data/raw/harper

python -m src.train --experiment projector_only
python -m src.train --experiment projector_head
python -m src.evaluate --checkpoint runs/projector_only/best.pt \
    --out results/projector_only
python -m src.evaluate --checkpoint runs/projector_head/best.pt --out results
python -m src.transcript_baseline

python -m src.train --experiment projector_only \
    --semantic-weight 0.05 --semantic-temperature 0.07 \
    --output-dir runs/projector_semantic
python -m src.semantic_eval --checkpoint runs/projector_semantic/best.pt --split test
python -m src.train --experiment projector_only --epochs 3 \
    --semantic-weight 0.05 --semantic-temperature 0.07 \
```

```

--output-dir runs/projector_semantic_3epochs
python -m src.semantic_eval \
--checkpoint runs/projector_semantic_3epochs/best.pt --split test

python -m src.transfer_eval --task configs/transfer_minds14.json \
--out results/transfer/minds14_zero_shot.json
python -m src.transfer_eval --task configs/transfer_urgency.json \
--skip-gold-transcript --out results/transfer/urgency_zero_shot.json
python -m src.transfer_eval --task configs/transfer_minds14.json \
--checkpoint-semantic runs/projector_semantic/best.pt \
--skip-text-baselines --out results/transfer/minds14_semantic_zero_shot.json
python -m src.transfer_eval --task configs/transfer_urgency.json \
--checkpoint-semantic runs/projector_semantic/best.pt \
--skip-text-baselines --skip-gold-transcript \
--out results/transfer/urgency_semantic_zero_shot.json
python -m src.transfer_eval --task configs/transfer_minds14.json \
--checkpoint-semantic runs/projector_semantic_3epochs/best.pt \
--skip-text-baselines --out results/transfer/minds14_semantic_3epochs.json
python -m src.transfer_eval --task configs/transfer_urgency.json \
--checkpoint-semantic runs/projector_semantic_3epochs/best.pt \
--skip-text-baselines --skip-gold-transcript \
--out results/transfer/urgency_semantic_3epochs.json
python -m unittest discover -s tests

```

The transfer commands evaluate A and B by default. The initial runs produce the text baselines; subsequent semantic-checkpoint runs skip those repeated baselines. Metrics, checkpoints, and partial run metadata are written under `results/` and `runs/`. Only the three-epoch semantic run has a recorded end-to-end time (492.11 seconds) and peak allocated device memory (2,406,236,160 bytes); project-wide compute and comparable timings for the other runs were not recorded.

3.3.4 Checklist summary

The statuses below follow the reporting questions in the cited checklist; they are not formal NeurIPS submission answers.

- **Code, data, and exact instructions — Partial:** scripts, configs, commands, and local results are available in the working copy; there is no public code revision or artifact URL, and raw audio/checkpoints are not bundled.
- **Experimental setting — Partial:** the data split and default hyperparameters are documented, but immutable model revisions and run-attested environment/configuration records are missing.
- **Statistical significance — No:** results are single-run point estimates without repeated seeds, confidence intervals, or error bars.
- **Compute resources — Partial:** the three-epoch semantic run records 492.11 seconds and peak allocated memory; comparable timings and the project-wide compute total are unavailable.
- **Assets and licenses — Partial:** dataset provenance and terms are cited, but model revisions and release terms must be confirmed before distributing weights or derived assets.

Reproducibility is therefore **partial**: the materials support informed reruns, but not an exact-reproduction claim.

Chapter 4

Results and discussion

4.1 Quantitative results

4.1.1 Robocall test split

A and B compare projector-only adaptation with adaptation that also updates Laya’s decision components; they do not hold the number of trainable parameters constant.

Table 4.1: Binary spam-versus-legitimate test results on the same 282 calls.

Method	Macro F1	AUROC
A: projector only	99.6%	100.0%
B: projector + decision head	100.0%	100.0%
C: Whisper ASR $\rightarrow$ Laya	89.2%	94.9%
A + semantic, 3 epochs	99.6%	100.0%
A + semantic, 10 epochs	100.0%	100.0%

These high scores show that the audio-to-decision pipeline can fit this held-out split. Because the split’s labels are completely confounded with source, they do not establish that it has learned robust, source-independent robocall cues. On this evaluation, recorded mean latency was 55.3 ms/call for A, 70.9 ms/call for B, and 762.3 ms/call for C (about 10.8–13.8 times slower for the ASR route). These are setup-specific per-call averages, not a general speed benchmark. In a supplementary 14-call normal-only role-play check, project notes report zero false alarms for A, B, and C; this sample is too small for a robust false-positive estimate, and its underlying data are not included or cleared for redistribution, so it is not used as headline evidence.

4.1.2 Zero-shot transfer

On MInDS-14, the direct-audio variants perform near the majority baseline and mostly predict `cash_deposit`.

The high C score comes from supplying text to Laya after ASR; it is evidence about that ASR-to-text-decision pipeline, not evidence that the direct-audio projector has learned 14-way intent recognition.

The reference-transcript result is an oracle-text control, not a matched baseline.

On urgency, B reaches 99.8% AUROC but detects only 9 of 20 urgent clips at the default decision threshold. The ten-epoch semantic model detects only 3 of 20; the three-epoch variant detects 16 of 20, but that result is exploratory because epoch count was chosen after seeing the earlier urgency evaluation on the same 40 clips. It must not be treated as an independent test result. C detects 19 of 20 urgent clips through the ASR-to-text route. All urgency results are especially uncertain because the dataset is small, synthetic, and may confound verbal content with vocal delivery.

Table 4.2: Transfer results. C uses transcript text, not the same input route as A/B.

Dataset	Method	Accuracy	Macro F1	Macro AUROC (OvR)
MInDS-14	Majority baseline	8.5%	1.1%	—
	A: projector only	8.0%	1.4%	49.7%
	B: projector + head	6.2%	2.9%	49.0%
	A + semantic, 10 epochs	8.5%	2.0%	50.8%
	A + semantic, 3 epochs*	6.6%	1.6%	49.8%
	C: Whisper ASR → Laya	93.6%	93.7%	99.3%
	C: reference transcript → Laya	90.9%	91.1%	99.0%
Urgency (40 clips)	Majority baseline	50.0%	33.3%	—
	A: projector only	45.0%	43.7%	51.8%
	B: projector + head	72.5%	70.3%	99.8%
	A + semantic, 10 epochs	52.5%	44.7%	72.8%
	A + semantic, 3 epochs*	80.0%	80.0%	88.8%
	C: Whisper ASR → Laya	97.5%	97.5%	100.0%

Figure 4.1 puts the three sets on one scale. It shows that the gap between the direct-audio and transcript routes is small in-domain but becomes the entire difference in the transfer setting, where the direct-audio variants land on the majority-class line rather than above it.

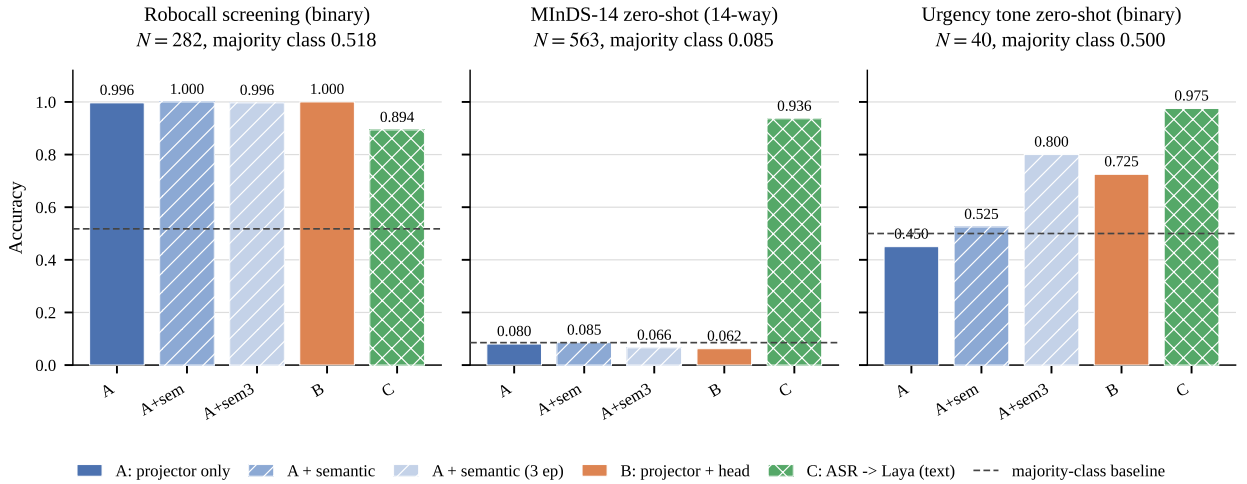


Figure 4.1: Accuracy of every variant on the three evaluation sets, plotted on a common 0–1 axis. In-domain robocall screening is at ceiling for the direct-audio variants, whereas zero-shot transfer to MInDS-14 collapses onto the majority-class baseline; the dashed line marks each set’s majority-class accuracy. Hatching marks route C, which consumes transcript text and is therefore not a matched input route for A and B. The three-epoch urgency bar is exploratory: its epoch count was chosen after an earlier urgency result on the same 40 clips.

4.1.3 What semantic alignment changed

The contrastive objective substantially improves audio-to-transcript retrieval on held-out FTC/Harper source data. It does not, by itself, turn the frozen spam decision head into a general intent classifier: the ten-epoch semantic model reaches only 8.5% accuracy and 2.0% Macro F1 on MInDS-14, and 52.5% accuracy on the urgency set. Semantic alignment and task-specific decision competence are distinct outcomes.

Table 4.3: Audio-to-transcript retrieval on 445 held-out source audio queries and 413 candidate texts. Random Recall@1 is about 0.24%.

Checkpoint	Retrieval loss	Recall@1	Recall@5
A, no semantic alignment	6.079	0.2%	1.1%
A + semantic, 3 epochs	4.041	16.4%	52.1%
A + semantic, 10 epochs	3.590	33.3%	74.8%

4.2 Interpretation

The results are most consistent with a **modular specialist-adaptation hypothesis**: a pretrained perception encoder can remain frozen while a trainable projector maps its features into a pretrained decision system. On this task, the projector-only variant updates 1.84 million parameters; adding the decision layers updates 28.09 million. The measured three-epoch adaptation run took just over eight minutes on one consumer GPU. At the portfolio level, the pattern is compatible with a System-1-style specialist architecture: modality-specific encoders and adapters can feed dedicated decision systems. This is a future architecture hypothesis; only the audio path was tested here.

This is not a claim that the whole deployed system is tiny: the Laya checkpoint alone has 421 million parameters according to its model card, and the Whisper encoder is also used. Nor does this experiment validate the same recipe for vision, sensors, or arbitrary audio tasks. Direct-audio transfer results are weak, and the main spam test is source-confounded. The evidence supports a promising engineering pattern that warrants cleaner experiments—not a general-purpose audio-understanding claim.

A typed decision interface could eventually return structured outputs such as:

```
{"spam": true, "risk": "high", "action": "block"}
```

This JSON object is an illustrative future design, not an output schema evaluated in this report. The experiments primarily evaluate the binary spam/legitimate choice. Laya emitted a calibration warning in this pipeline; its reported scores must not be presented as calibrated confidence.

Chapter 5

Limitations, responsible use, and conclusions

5.1 Limitations and responsible use

The results should be interpreted within the following limits:

1. **Source confounding:** the internal spam test maps source to label exactly; a source-independent test split is required.
2. **Small or non-independent transfer sets:** MInDS-14 lacks an official held-out test split; urgency uses 40 synthetic clips; the three-epoch urgency result is post hoc.
3. **Route mismatch:** A/B process audio directly, whereas C first transcribes it. Their scores answer different pipeline questions.
4. **Base-model prior:** Laya is already a pretrained text decision model. This study does not separate audio alignment from prior text-task competence.
5. **Model size and compute:** reported trainable parameters and one training time do not describe total inference memory, energy, or cost. The ten-epoch runtime was not measured; per-call latency also depends on hardware and audio duration.
6. **Uncertainty and variant fairness:** results are single-run point estimates without repeated seeds or confidence intervals. B updates about 15 times as many parameters as A, so the A/B comparison is not a parameter-matched ablation and small score differences should not be read as statistically meaningful.
7. **Incomplete run provenance:** model weights are not revision-pinned, dependencies are not locked, the source tree has no recorded Git revision, and historical run metadata omit the complete environment and effective configuration. Re-run commands are available, but exact reproduction is not guaranteed.
8. **No cross-modality evidence:** no image, video, or sensor input was tested.
9. **Deployment:** this is a research prototype, not a validated call-blocking system. Do not use its uncalibrated scores for consequential decisions without independent validation, threshold selection, and error analysis.

The most informative next experiment is a larger, diverse paired speech–transcript corpus with speaker- and corpus-held-out evaluation. The decision head should also be evaluated on independently labeled task sets, while keeping final test labels untouched during checkpoint and epoch selection. Human-recorded urgency speech and explicit false-positive/false-negative costs are needed before any operational claim. Other modalities should be tested as separate experiments rather than inferred from the audio results.

5.2 Conclusion

Audio-Laya provides an initial test of a reusable adaptation recipe: freeze a pretrained modality encoder, train a lightweight projector, and optionally update a task-specific decision head. The internal split demonstrates that this arrangement can achieve a high score on a defined audio decision task, but source confounding prevents a robust spam-generalization claim. Transfer tests and semantic alignment further show that retrieving transcript meaning does not automatically produce new-task decision competence. The defensible conclusion is therefore **specialized adaptation is promising; universal generalization remains unproven.**

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